

HRMS Network-Based Attendance System

Simple Project Documentation

1. Introduction

This project is a web-based HRMS login system that allows employees to log in only when connected to the organization's authorized office network. The system uses a local Network Agent to collect network information and verifies it through the backend before allowing authentication.

Objectives

- Secure employee login
- Office network verification
- JWT authentication
- Prevent unauthorized access

Technology Stack

Frontend: React + TypeScript + Vite

Backend: Node.js + Express.js

Database: MongoDB Atlas

Authentication: JWT + bcrypt

Network Agent: Node.js

2. System Architecture

Employee → React Frontend → Network Agent (localhost:5001) → Backend API → MongoDB →
JWT Authentication → Dashboard

Modules

Frontend: Login, Signup, Dashboard, Network Status Card.

Backend: Authentication, Network Verification, User Management.

Network Agent: Reads SSID, BSSID, Gateway and Subnet.

3. Backend Structure

```
backend/  
server.js  
src/config/database.js  
controllers/auth.controller.js  
middleware/auth.middleware.js  
models/User.js, OfficeNetwork.js  
routes/auth.routes.js  
services/auth.service.js, network.service.js
```

Main Responsibilities

auth.controller.js handles login and signup. auth.service.js validates users and hashes passwords.
network.service.js verifies whether the current network matches the authorized office network stored in MongoDB.

4. Frontend & Network Agent

```
frontend/  
pages/Login.tsx  
Signup.tsx  
Dashboard.tsx
```

components/NetworkStatusCard.tsx
services/api.ts

The Network Agent runs locally and exposes network information through `http://localhost:5001/network`. The frontend reads this information before requesting login.

5. Login Flow

1. Employee opens Login page.
2. Frontend requests Network Agent.
3. Network Agent returns network details.
4. Backend verifies office network.
5. If authorized, employee credentials are validated.
6. JWT token is generated.
7. Employee is redirected to the Dashboard.

Security Features

JWT Authentication, bcrypt password hashing, protected routes, office network verification, MongoDB data storage.

Conclusion

The project provides a secure login mechanism by combining employee authentication with office network verification. Its modular architecture separates the frontend, backend, and Network Agent, making the application scalable and maintainable.